



CS & IT ENGINEERING

Discrete Mathematics

Mathematical Logic

Lecture No.- 04



By- Satish Sir

Recap of Previous Lecture



Topic

Logic Part-03



Topics to be Covered



Topic

Logic Part-04



Type-3: Inference Rule:

premises → conclusion.
True check.

Consider premises as True.
Check the conclusion.
if conclusion is True.
whole expression comes
under Inference Rule.

IS ✓

valid.

True.

False

premises → conclusion.
True True.

IS ✗

premises → conclusion.
True False

valid/tautology.

$$\underline{a \wedge b \rightarrow b}$$

check the validity



2 cases w.r.t to the left side.



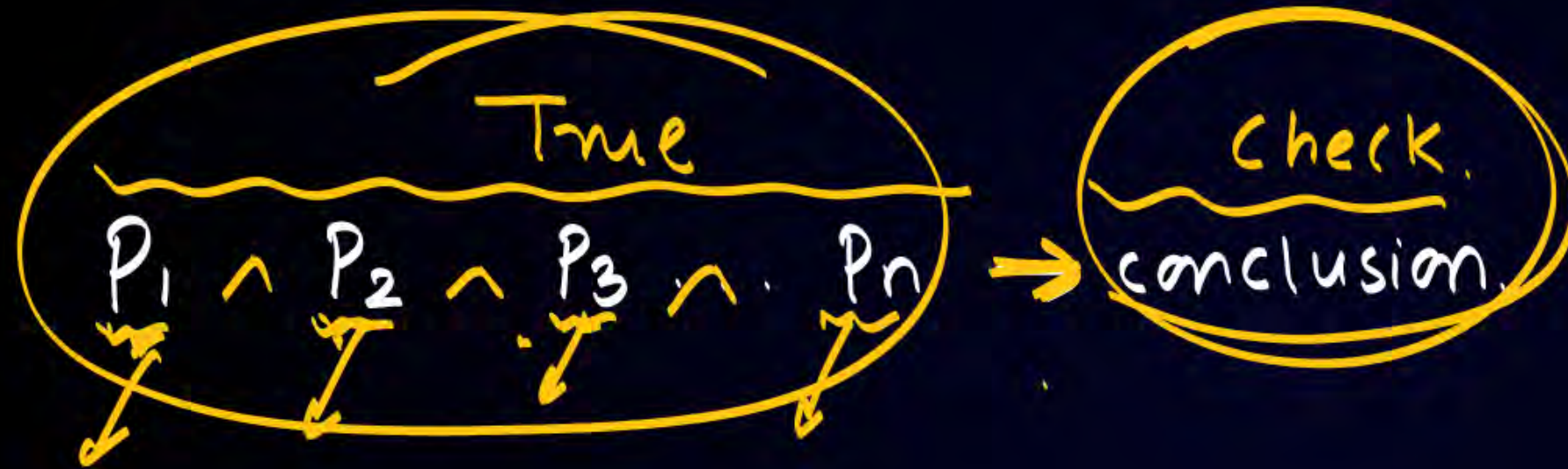
arguments:



premises $\rightarrow$ conclusion.

OR

premises.
 $\therefore$ conclusion



OR

premises 1.
 $\wedge$
premises 2.
 $\wedge$
premises 3.
 $\vdots$

Conclusion.

I.R.

premises 1 : mobile phone either it is left or Right pocket (True)

premises 2: mobile phone it is not in left pocket (True)

Conclusion: mobile phone it is in Right pocket.

$$P_1: L \vee R (T)$$

$$P_2: \neg L (T)$$

$$Q: R$$

$$P_1 \wedge P_2 \rightarrow Q$$

$$((L \vee R) \wedge \neg L) \rightarrow R$$

check the validity.

$$((L \vee R) \wedge \neg L) \rightarrow R$$

Tautology/valid.

L	R	$\neg L$	$L \vee R$
T	T	F	T
T	F	F	T
F	T	T	T
F	F	T	F

Type-1:

$$\frac{T}{((L \vee R) \wedge \neg L) \rightarrow R} \frac{F}{R}$$

True

$$R = F$$

$$L = T$$

$$\neg L = F$$



P_1 : if $\overset{P}{\boxed{\text{p.m. exist}}}$ then $\overset{Q}{\boxed{\text{no. of vertices will be even}}}$. (True)

P_2 : $\boxed{\text{p.m. exist}}$ $\overset{P}{}$

Q : no. of vertices will be even.

$$\frac{P \rightarrow Q}{P}$$
 OR

$$\therefore Q$$

$(\overset{P}{\boxed{P}} \overset{Q}{\boxed{(P \rightarrow Q)}} \rightarrow Q$ ✓

P	Q	$P \rightarrow Q$
T	T	T
T	F	F
F	T	T
F	F	T

P_1 : if $\overset{P}{\text{p.m exist}}$ then $\overset{Q}{\text{no. of vertices}}$ are even. (True)

P_2 : no. of vertices are even. (True)
(Q)

$Q \therefore$ p.m exist

$$((P \rightarrow Q) \wedge Q) \rightarrow P$$

$$\frac{P \rightarrow Q \quad Q}{\therefore P}$$

P	Q	$P \rightarrow Q$
T	T	T
T	F	F
F	T	T
F	F	T

$$1) \frac{\frac{p \rightarrow q}{p}}{q} \quad \frac{(p \rightarrow q) \wedge p}{(modus ponens)} \rightarrow q$$

$$2) \frac{p \rightarrow q}{\frac{\neg q}{\neg p}} \equiv \frac{\neg q}{\neg p} \quad (modus tollens)$$

$$3) \frac{p \rightarrow q}{Q \rightarrow R} \quad \frac{((p \rightarrow q) \wedge (Q \rightarrow R)) \rightarrow (p \rightarrow R)}{(Hypothetical syllogism)}$$

$$4) \frac{p \vee q}{\neg p} \quad \frac{((p \vee q) \wedge \neg p) \rightarrow q}{(Disjunctive syllogism)}$$

$$5) \frac{p}{p \vee q} \quad (Addition) \quad \frac{p}{T} \rightarrow \frac{p \vee q}{T \vee}$$

$$\frac{p \rightarrow q}{Q} \quad p$$

Fallacy

$$6) \frac{p \wedge q}{p} \quad \text{OR} \quad \frac{T}{p \wedge q \rightarrow p} \quad (simplification)$$

$$7) \frac{p \vee q}{\neg p \vee R} \quad \frac{T}{((p \vee q) \wedge (\neg p \vee R)) \rightarrow (Q \vee R)} \quad (Resolution)$$

Check the validity?

$$1) (p \wedge (p \rightarrow q)) \rightarrow q$$

$$\begin{array}{l} p \\ p \rightarrow q \\ \hline q \end{array} \quad (\text{modus ponens})$$

$$2) (p \wedge (p \rightarrow q) \wedge (q \rightarrow m)) \rightarrow m$$

$$\begin{array}{l} 1. p \\ 2. p \rightarrow q \end{array} \quad (\text{modus ponens})$$

$$\begin{array}{l} q \\ q \rightarrow m \\ \hline m \end{array} \quad (\text{modus ponens})$$

$$3) \quad \underbrace{(m \wedge (m \rightarrow n) \wedge (\neg n \vee a))}_{\text{method 2}} \rightarrow a$$

method 2:

$$\begin{array}{l} 1. \quad m \\ 2. \quad m \rightarrow n \\ \hline n. \\ 3. \quad \neg n \vee a. \quad (D.S) \\ \hline a. \end{array}$$

$$\begin{array}{l} m. \\ m \rightarrow n. \\ \hline n. \\ 3. \quad \neg n \vee a. \equiv \underline{n \rightarrow a.} \\ \hline a. \text{ (valid)} \end{array}$$

$\equiv$
 $\rightarrow$

$$\left((a \rightarrow b) \wedge (b \rightarrow c) \wedge (a \vee \neg x) \right) \rightarrow \underline{\underline{\neg x \vee c.}}$$

$$\begin{array}{l} a \rightarrow b \\ b \rightarrow c. \\ \hline a \rightarrow c. \equiv \underline{\neg a \vee c.} \\ 3. \ a \vee \neg x. \quad \underline{a \vee \neg x.} \quad \text{(Resolution)} \\ \hline c \vee \neg x. \quad \checkmark \end{array}$$

{ every Type-3.
we can solve
with Type-2

$$\left((p \rightarrow q) \wedge (\neg R \vee s) \wedge (p \vee R) \right) \rightarrow (\neg q \rightarrow s)$$

$q \vee s$

$$\begin{array}{l} p \rightarrow q \quad \rightarrow \quad \textcircled{\neg p \vee q} \\ \textcircled{\neg R \vee s} \quad \rightarrow \quad \textcircled{p \vee s} \\ p \vee \textcircled{R} \quad \rightarrow \quad \textcircled{q \vee s} \end{array}$$

$$\begin{array}{l} \text{c)} \quad p \rightarrow q \\ \quad \neg q \\ \quad \neg r \\ \hline \therefore \neg(p \vee r) \end{array}$$

$$\begin{array}{l} \text{e)} \quad p \rightarrow (q \rightarrow r) \\ \quad \neg q \rightarrow \neg p \\ \quad p \\ \hline \therefore r \end{array}$$

$$\begin{array}{l} \text{g)} \quad p \rightarrow (q \rightarrow r) \\ \quad p \vee s \\ \quad t \rightarrow q \\ \quad \neg s \\ \hline \therefore \neg r \rightarrow \neg t \end{array}$$

$$\begin{array}{l} \text{d)} \quad p \rightarrow q \\ \quad r \rightarrow \neg q \\ \quad r \\ \hline \therefore \neg p \end{array}$$

$$\begin{array}{l} \text{f)} \quad p \wedge q \\ \quad p \rightarrow (r \wedge q) \\ \quad r \rightarrow (s \vee t) \\ \quad \neg s \\ \hline \therefore t \end{array}$$

$$\begin{array}{l} \text{h)} \quad p \vee q \\ \quad \neg p \vee r \\ \quad \neg r \\ \hline \therefore q \end{array}$$

a) $[(p \wedge \neg q) \wedge [p \rightarrow (q \rightarrow r)]] \rightarrow \neg r$

b) $[[(p \wedge q) \rightarrow r] \wedge (\neg q \vee r)] \rightarrow p$

c)

$$\begin{array}{l} p \leftrightarrow q \\ q \rightarrow r \\ r \vee \neg s \\ \neg s \rightarrow q \\ \hline \therefore s \end{array}$$

d)

$$\begin{array}{l} p \\ p \rightarrow r \\ p \rightarrow (q \vee \neg r) \\ \neg q \vee \neg s \\ \hline \therefore s \end{array}$$

$$\begin{array}{l}
 \text{(i)} \quad p \vee (q \wedge r) \\
 \quad \quad p \rightarrow s \\
 \hline
 \therefore r \vee s
 \end{array}$$

$$\begin{array}{l}
 \text{(iii)} \quad p \vee q \\
 \quad \quad p \rightarrow r \\
 \quad \quad r \rightarrow s \\
 \hline
 \therefore q \vee s
 \end{array}$$

$$\begin{array}{l}
 \text{(v)} \quad \neg p \vee s \\
 \quad \quad \neg t \vee (s \wedge r) \\
 \quad \quad \neg q \vee r \\
 \quad \quad p \vee q \vee t \\
 \hline
 \therefore r \vee s
 \end{array}$$

$$\begin{array}{l}
 \text{(ii)} \quad p \\
 \quad \quad p \leftrightarrow q \\
 \hline
 \therefore q
 \end{array}$$

$$\begin{array}{l}
 \text{(iv)} \quad \neg p \vee q \vee r \\
 \quad \quad \neg q \\
 \quad \quad \neg r \\
 \hline
 \therefore \neg p
 \end{array}$$

$$\begin{array}{l}
 p \\
 p \vee q \\
 q \rightarrow (r \rightarrow s) \\
 t \rightarrow r \\
 \hline
 \therefore \neg s \rightarrow \neg t
 \end{array}$$

$$\begin{array}{l}
 u \rightarrow r \\
 (r \wedge s) \rightarrow (p \vee t) \\
 q \rightarrow (u \wedge s) \\
 \neg t \\
 \hline
 \therefore q \rightarrow p
 \end{array}$$

$$\begin{array}{l}
 p \rightarrow q \\
 q \rightarrow (r \wedge s) \\
 \neg r \vee (\neg t \vee u) \\
 p \wedge t \\
 \hline
 \therefore u
 \end{array}$$



2 mins Summary

Topic

One

Logic Part-04

Topic

Two

Topic

Three

Topic

Four

Topic

Five



THANK - YOU